

CBSE Class 12 Chemistry — Biomolecules

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQs / Assertion-Reason / Very Short Answer)

Q1. Which of the following is a polysaccharide? (a) Glucose (b) Maltose (c) Glycogen (d) Lactose [PYQ SQP Term-1 2021] [Confidence: Very High]

Q2. Which one of the following reactions is not explained by the open-chain structure of glucose? (a) Formation of pentaacetate of glucose with acetic anhydride. (b) Formation of addition product with 2,4-DNP reagent. (c) Silver mirror formation with Tollen's reagent. (d) Reaction with **HI** to form n-hexane. [PYQ SQP Term-1 2021] [Confidence: Very High]

Q3. Glucose on reaction with **Br₂** water gives: (a) saccharic acid (b) hexanoic acid (c) gluconic acid (d) salicylic acid [PYQ Term-1 2021] [Confidence: Very High]

Q4. Which of the following sugars is known as dextrose? (a) Glucose (b) Fructose (c) Ribose (d) Sucrose [PYQ Term-1 2021] [Confidence: High]

Q5. Find the incorrect statement from the following for glucose: (a) It is present in furanose form. (b) It is an aldohexose. (c) It does not give 2,4-DNP test. (d) It forms n-hexane when heated with **HI**. [PYQ Delhi Gov. SQP Term-1 2021] [Confidence: Very High]

Q6. The vitamins which can be stored in our body are: (a) Vitamin A, B, D and E (b) Vitamin A, C, D and K (c) Vitamin A, B, C and D (d) Vitamin A, D, E and K [PYQ SQP 2023] [Confidence: High]

Q7. Complete the following analogy: Curdling of milk : A :: α -helix : B (a) A: Primary structure, B : Secondary structure (b) A: Denatured protein, B : Primary structure (c) A: Secondary structure, B : Denatured protein (d) A: Denatured protein, B : Secondary structure [PYQ Term-1 2021] [Confidence: Very High]

Q8. Peptide linkage is present in: (a) carbohydrates (b) vitamins (c) protein (d) rubber [PYQ 2020] [Confidence: Very High]

Q9. Amino acids are: (a) acidic (b) basic (c) amphoteric (d) neutral [PYQ 2020] [Confidence: Very High]

Q10. Nucleosides are composed of: (a) a pentose sugar and phosphoric acid. (b) a nitrogenous base and phosphoric acid. (c) a nitrogenous base and a pentose sugar. (d) a nitrogenous base, a pentose sugar and phosphoric acid. [PYQ Term-1 2021] [Confidence: Very High]

Q11. The base which is present in DNA but not in RNA, is: (a) Cytosine (b) Guanine (c) Adenine (d) Thymine [PYQ Term-1 2021] [Confidence: Very High]

Q12. Assertion (A): Sucrose is a non-reducing sugar. Reason (R): It has glycosidic linkage. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ 2020] [Confidence: Very High]

Q13. Assertion (A): Globular proteins are generally soluble in water. Reason (R): In globular proteins, the polypeptide chain folds around to give a spherical shape. (a) Both A and R are true and R is the

correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ SQP 2022] [Confidence: High]

Q14. Name the species formed when an aqueous solution of an amino acid is dissolved in water. [PYQ SQP 2019] [Confidence: Very High]

Q15. Which of the two components of starch is water-soluble? [PYQ 2014] [Confidence: Very High]

Q16. Write the name of the linkage joining two monosaccharides. [PYQ 2020] [Confidence: Very High]

Q17. Name the disaccharide which on hydrolysis gives two molecules of glucose. [PYQ 2020] [Confidence: High]

Q18. Which one of the following is a monosaccharide: Starch, Maltose, Fructose, Cellulose? [PYQ 2015] [Confidence: Very High]

Q19. Deficiency of which vitamin causes scurvy? [PYQ 2014] [Confidence: Very High]

Q20. Name the base that is found in the nucleotide of RNA only. [PYQ 2014] [Confidence: Very High]

2-Mark Questions (Short Answer)

Q21. Differentiate between the following: (A) Amylose and Amylopectin (B) Peptide linkage and glycosidic linkage [PYQ 2019] [Confidence: Very High]

Q22. Define the following terms: (A) Oligosaccharides (B) Invert sugar [PYQ 2020] [Confidence: High]

Q23. What happens when D-glucose is treated with the following reagents? (A) Bromine water (B) HNO_3 [PYQ SQP 2022] [Confidence: Very High]

Q24. Account for the following: (A) There are 5 $-\text{OH}$ groups in glucose. (B) Glucose is a reducing sugar. [PYQ SQP 2022] [Confidence: High]

Q25. Write the products of hydrolysis of lactose. [PYQ 2019] [Confidence: Very High]

Q26. What is the difference between fibrous protein and globular protein? [PYQ 2017, 2015] [Confidence: Very High]

Q27. Give one example each for fibrous and globular protein. [PYQ 2016] [Confidence: High]

Q28. Define the following with an example of each: (A) Denatured Protein. (B) Essential Amino Acid. [PYQ 2018] [Confidence: Very High]

Q29. What is the difference between acidic amino acids and basic amino acids? [PYQ 2015] [Confidence: Medium]

Q30. What is the difference between a nucleotide and nucleoside? [PYQ 2016, 2014] [Confidence: Very High]

Q31. Why Vitamin C can't be stored in our body? [PYQ 2016] [Confidence: Very High]

Q32. Name the vitamin whose deficiency causes bone deformities in children. Write its one dietary source. [PYQ 2015 / Practice] [Confidence: High]

3-Mark Questions (Short Answer)

- Q33.** Write chemical reactions to show that the open structure of D-glucose contains the following: (A) Straight chain (B) Five alcohol groups (C) Aldehyde as carbonyl group [PYQ 2019] [Confidence: Very High]
- Q34.** Deficiency of which vitamin causes: (A) night blindness (B) poor coagulation of blood (C) scurvy [PYQ 2014] [Confidence: Very High]
- Q35.** Account for the following: (A) Amino acids behave like salts rather than simple amines or carboxylic acids. (B) The two strands of DNA are complementary to each other. (C) Reaction of D-glucose with $H_2N - OH$. [PYQ 2015 / Practice] [Confidence: Very High]
- Q36.** Explain what is meant by: (A) primary structure of proteins (B) secondary structure of proteins (C) tertiary structure of proteins [Practice] [Confidence: High]
- Q37.** Write the exact products obtained when D-glucose reacts with: (A) $H_2N - OH$ (B) HCN (C) HI (heated) [PYQ 2015 / Practice] [Confidence: Very High]
- Q38.** Differentiate between DNA and RNA (any three points of difference). [Practice] [Confidence: Very High]
- Q39.** Answer the following: (A) Which structural change will occur when a native protein is subjected to a change in pH? (B) Name the linkage present in nucleic acids. (C) Why are vitamin A and vitamin C essential for us? [PYQ SQP 2023 / 2016 / 2014] [Confidence: High]
- Q40.** Distinguish between: (A) Acidic and basic amino acids. (B) Reducing and non-reducing sugars. (C) α -helix and β -pleated sheet structures. [PYQ 2015 / Practice] [Confidence: High]
- Q41.** Answer the following: (A) Which of the two components of starch is water-soluble? (B) Name the base found in RNA only. (C) What are polysaccharides? [PYQ 2014] [Confidence: Very High]
- Q42.** Glucose on reaction with acetic anhydride gives glucose pentaacetate. What does it suggest about the structure of glucose? How is it different from the reaction of glucose with HNO_3 ? [PYQ 2017 / Practice] [Confidence: Medium]
-

5-Mark Questions (Long Answer / Case-Study)

- Q43. Case Study: Inversion of Cane Sugar** When sucrose is hydrolysed, the optical rotation values are measured using a polarimeter and are given in the following table: Time = 0 hours, Specific Rotation = $+66.5^\circ$ Time = ∞ hours, Specific Rotation = -39.9° (A) Account for the two specific rotation values. (2 Marks) (B) What is the specific name given to sucrose based on the above observation? (1 Mark) (C) One of the products formed during the hydrolysis of sucrose is glucose, which reacts with hydroxylamine to give compound A. Identify compound A. (1 Mark) (D) Name the two monosaccharides that form sucrose. (1 Mark) [PYQ SQP 2023 / Practice] [Confidence: Very High]
- Q44. Case Study: Proteins and their Structure** Proteins are polymers of amino acids linked by peptide bonds. They exhibit complex structural organizations classified into primary, secondary, tertiary, and quaternary structures. Denaturation occurs when a protein loses its native shape due to heat or chemical changes. (A) What is the effect of denaturation on the primary, secondary, and tertiary structures of proteins? (2 Marks) (B) Differentiate between essential and non-essential amino acids. (1 Mark) (C) Draw the zwitterion structure of an amino acid. (1 Mark) (D) Name the two major types of secondary structures found in proteins. (1 Mark) [Practice / PYQ 2018 Themes] [Confidence: Very High]

Q45. Case Study: Nucleic Acids and DNA DNA and RNA are nucleic acids responsible for the storage and transmission of genetic information. They are composed of repeating units called nucleotides. (A) Why are the two strands of DNA not identical but complementary? (2 Marks) (B) Name the pentose sugar present in DNA and RNA respectively. (1 Mark) (C) What is DNA fingerprinting and what is its primary application? (1 Mark) (D) What type of forces hold the two strands of DNA together? (1 Mark) [PYQ SQP 2022 / Practice] [Confidence: Very High]

Q46. (A) Write the product when D-glucose reacts with conc. HNO_3 . (1 Mark) (B) Glucose on reaction with acetic anhydride gives glucose pentaacetate. What does it suggest about the structure of glucose? (1 Mark) (C) Name the vitamin whose deficiency causes scurvy. (1 Mark) (D) Give one example each of a water-soluble vitamin and a fat-soluble vitamin. (2 Marks) [PYQ 2018, 2017, 2014] [Confidence: High]

Q47. (A) Define Mutarotation. (1 Mark) (B) Draw the Haworth structures of α -D-(+)-glucopyranose and β -D-(+)-glucopyranose. (2 Marks) (C) Why does sucrose not show mutarotation? (1 Mark) (D) What are Zwitter ions? (1 Mark) [Practice] [Confidence: Medium]

Q48. Carbohydrates and their Chemistry (A) Define carbohydrates. (1 Mark) (B) Classify carbohydrates on the basis of their behaviour on hydrolysis, giving one example of each. (2 Marks) (C) How do you test for a reducing sugar? (1 Mark) (D) Why is cellulose not digestible by humans? (1 Mark) [Practice] [Confidence: High]

Q49. (A) What is the fundamental difference between nucleosides and nucleotides? (2 Marks) (B) Describe the Watson-Crick double helix model of DNA briefly. (1 Mark) (C) List two important biological functions of nucleic acids. (1 Mark) (D) Which nitrogenous base is replaced by uracil in RNA? (1 Mark) [PYQ 2016 / Practice] [Confidence: Very High]

Q50. (A) Explain the term 'invert sugar'. (1 Mark) (B) Name the forces responsible for the secondary and tertiary structures of proteins. (2 Marks) (C) Why are some amino acids called 'essential' amino acids? (1 Mark) (D) Write the exact chemical reaction showing that glucose contains a straight chain of six carbon atoms. (1 Mark) [Practice] [Confidence: Very High]

Answer Key

Q1. (c) Glycogen. **Q2.** (b) Formation of addition product with 2,4-DNP reagent. (Open chain cannot explain absence of 2,4-DNP test). **Q3.** (c) gluconic acid. **Q4.** (a) Glucose. **Q5.** (a) It is present in furanose form. (Incorrect, it is present in pyranose form). **Q6.** (d) Vitamin A, D, E and K. (Fat-soluble vitamins can be stored). **Q7.** (d) A: Denatured protein, B : Secondary structure. **Q8.** (c) protein. **Q9.** (c) amphoteric. (Due to the presence of both acidic $-COOH$ and basic $-NH_2$ groups forming a zwitterion). **Q10.** (b) a nitrogenous base and a pentose sugar. **Q11.** (d) Thymine. **Q12.** (b) Both A and R are true but R is not the correct explanation of A. (Non-reducing because the reducing groups of both glucose and fructose are involved in glycosidic bond formation). **Q13.** (a) Both A and R are true and R is the correct explanation of A. **Q14.** Zwitterion (dipolar ion). **Q15.** Amylose. **Q16.** Glycosidic linkage. **Q17.** Maltose. **Q18.** Fructose. **Q19.** Vitamin C. **Q20.** Uracil.

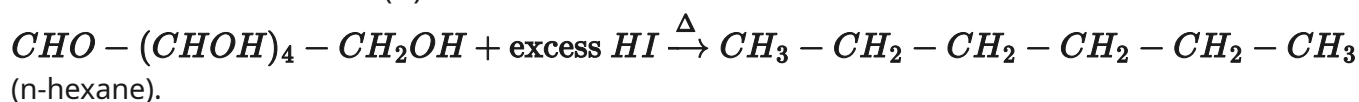
Q21. (A) Amylose is a water-soluble, unbranched polymer of α -D-glucose with C1-C4 linkages. Amylopectin is a water-insoluble, branched polymer with C1-C4 and C1-C6 linkages. (B) Peptide linkage is an amide bond ($-CO-NH-$) between amino acids in proteins. Glycosidic linkage is

an oxide linkage ($—O—$) joining two monosaccharides in carbohydrates. **Q22.** (A) Carbohydrates that yield 2 to 10 monosaccharide units on hydrolysis. (B) The equimolar mixture of glucose and fructose obtained by the hydrolysis of sucrose is called invert sugar because the sign of optical rotation inverts from dextro to laevo. **Q23.** (A) Gluconic acid is formed (oxidation of aldehyde group). (B) Saccharic acid is formed (oxidation of both aldehyde and primary alcohol groups). **Q24.** (A) Acetylation of glucose with acetic anhydride yields glucose pentaacetate, indicating the presence of 5 $—OH$ groups. (B) Glucose reduces Tollen's and Fehling's reagents because it has a free aldehydic group in its open-chain structure (or hemiacetal form). **Q25.** D-Galactose and D-Glucose. **Q26.** Fibrous proteins are thread-like, water-insoluble, and have strong intermolecular forces (hydrogen & disulphide bonds). Globular proteins have a spherical shape, are water-soluble, and polypeptide chains are folded around. **Q27.** Fibrous: Keratin / Myosin / Collagen. Globular: Insulin / Albumin. **Q28.** (A) When a native protein is subjected to heat or chemical change, its secondary and tertiary structures unfold, losing biological activity. (B) Amino acids that cannot be synthesized in the body and must be obtained through diet (e.g., Valine, Leucine). **Q29.** Acidic amino acids contain more carboxyl groups than amino groups (e.g., Aspartic acid). Basic amino acids contain more amino groups than carboxyl groups (e.g., Lysine). **Q30.** A nucleoside contains a pentose sugar and a nitrogenous base. A nucleotide contains a pentose sugar, a nitrogenous base, and a phosphate group. **Q31.** Vitamin C is water-soluble. It is readily excreted in urine and cannot be stored in our body. **Q32.** Vitamin D. Dietary source: Milk, egg yolk, fish liver oil, or sunlight exposure.

Q33. (A) Prolonged heating with HI gives n-hexane. (B) Acetylation with acetic anhydride gives glucose pentaacetate. (C) Reaction with hydroxylamine ($H_2N—OH$) gives an oxime. **Q34.** (A) In aqueous solution, the carboxyl group loses a proton and the amino group accepts it, forming a dipolar Zwitterion. (B) Hydrogen bonding strictly pairs specific bases: Adenine forms 2 H-bonds with Thymine, and Guanine forms 3 H-bonds with Cytosine. (C) Glucose reacts with hydroxylamine to form glucose oxime ($CH=N—OH$). **Q35.** (A) Vitamin A. (B) Vitamin K. (C) Vitamin C. **Q36.** (A) The specific sequence in which amino acids are linked in a polypeptide. (B) The shape in which a long polypeptide chain exists (e.g., α -helix, β -pleated sheet) due to H-bonding. (C) The overall folding of the entire polypeptide chain, giving a 3D shape. **Q37.** (A) Glucose oxime. (B) Glucose cyanohydrin. (C) n-hexane. **Q38.** DNA: Deoxyribose sugar, contains Thymine, double-stranded, stores genetic info. RNA: Ribose sugar, contains Uracil, single-stranded, synthesizes proteins. **Q39.** (A) Denaturation occurs; secondary and tertiary structures are destroyed but the primary structure remains intact. (B) Phosphodiester linkage. (C) Vitamin A is essential for vision/eyesight. Vitamin C is essential to prevent scurvy and build immunity. They cannot be synthesized by the body. **Q40.** (A) Acidic: Contains more $—COOH$ than $—NH_2$ groups. Basic: Contains more $—NH_2$ than $—COOH$ groups. (B) Reducing sugars reduce Tollen's/Fehling's reagent (free hemiacetal/anomeric carbon). Non-reducing sugars do not (anomeric carbons are locked in glycosidic bonds). (C) α -helix: coiled structure stabilized by intramolecular H-bonds. β -pleated sheet: stretched chains laid side by side stabilized by intermolecular H-bonds. **Q41.** (A) Amylose. (B) Uracil. (C) Carbohydrates that yield a large number of monosaccharide units on hydrolysis (e.g., starch, cellulose). **Q42.** Pentaacetate formation confirms the presence of 5 hydroxyl ($—OH$) groups attached to different carbon atoms. With HNO_3 , it oxidizes to saccharic acid (a dicarboxylic acid), proving the presence of a primary alcohol group along with the aldehyde group.

Q43. (A) Initial $+66.5^\circ$ is the rotation of pure sucrose (dextrorotatory). After hydrolysis, an equimolar mixture of D-glucose ($+52.5^\circ$) and D-fructose (-92.4°) forms. Since laevorotation

exceeds dextrorotation, the net rotation is negative (-39.9°). (B) Invert Sugar. (C) Glucose oxime. (D) α -D-Glucose and β -D-Fructose. **Q44.** (A) Secondary and tertiary structures are destroyed (unfolded) due to the breaking of H-bonds, but the primary structure (peptide bonds) remains intact. (B) Essential: Must be acquired through diet. Non-essential: Synthesized by the body. (C) $H_3N^+ - CH(R) - COO^-$. (D) α -helix and β -pleated sheet. **Q45.** (A) They are held together by specific hydrogen bonding between purine and pyrimidine bases: Adenine always pairs with Thymine, and Guanine always pairs with Cytosine. Thus, the sequence of one strand dictates the other. (B) DNA: β -D-2-deoxyribose. RNA: β -D-ribose. (C) The unique sequence of bases in an individual's DNA. Application: Forensic science (paternity testing, criminal identification). (D) Hydrogen bonds. **Q46.** (A) Saccharic acid. (B) It confirms the presence of five hydroxyl ($-OH$) groups. (C) Vitamin C (Ascorbic acid). (D) Water-soluble: Vitamin C (or B). Fat-soluble: Vitamin A (or D, E, K). **Q47.** (A) The spontaneous change in specific optical rotation of an optically active compound with time to an equilibrium value. (B) Draw six-membered pyranose rings (for α , $-OH$ on C1 is below the plane; for β , $-OH$ on C1 is above the plane). (C) Sucrose has no free hemiacetal/anomeric carbon to open into the chain form; both are involved in the glycosidic bond. (D) Dipolar ions formed when the carboxyl group loses a proton and the amino group accepts it in an amino acid. **Q48.** (A) Polyhydroxy aldehydes or ketones, or compounds that produce them on hydrolysis. (B) Monosaccharides (cannot be hydrolyzed, e.g., Glucose). Oligosaccharides (yield 2-10 units, e.g., Sucrose). Polysaccharides (yield many units, e.g., Starch). (C) Boil with Fehling's or Tollen's reagent. A reducing sugar will give a red precipitate or silver mirror. (D) Humans lack the enzyme cellulase needed to break the β -glycosidic linkages in cellulose. **Q49.** (A) Nucleoside = Base + Sugar. Nucleotide = Base + Sugar + Phosphate. (B) Two polynucleotide chains wound in a right-handed double helix, held together by hydrogen bonds between specific complementary base pairs ($A=T$, $G\equiv C$) pointing inward. (C) 1. Storage/transmission of genetic information (heredity). 2. Protein synthesis. (D) Thymine. **Q50.** (A) The equimolar mixture of glucose and fructose formed by sucrose hydrolysis, changing optical rotation from dextro to laevo. (B) Hydrogen bonds, disulphide linkages, van der Waals, and electrostatic forces. (C) Because they cannot be synthesized by the human body and must be taken via diet. (D)



CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).